

Linear Regression Evaluation Across Vectorized and Non-Vectorized Techniques

1 Introduction

Linear regression is one of the most fundamental supervised learning algorithms used for predicting continuous values. In this assignment, two implementations of linear regression using gradient descent are developed and analyzed:

- A loop-based implementation using explicit iteration over data samples
- A vectorized implementation using matrix operations in NumPy

The primary objective is to evaluate both implementations in terms of convergence behavior, computational efficiency, and predictive performance.

2 Objective

The goal of this experiment is to:

- Compare convergence speed between loop-based and vectorized implementations
- Measure and analyze computational efficiency using training time
- Evaluate predictive performance using MAE, RMSE, and R^2 score
- Visualize convergence and performance differences between both methods

3 Dataset and Preprocessing

The dataset used contains housing-related features such as number of rooms, distance, number of bathrooms, car spaces, land size, building area, and bedroom count. The target variable is the house price.

Preprocessing steps include:

- Removal of missing values
- Feature selection from raw dataset
- Train-test split (80% training, 20% testing)
- Feature normalization using training mean and standard deviation
- Standardization of target variable for stable optimization
- Addition of bias term to input features

4 Methodology

4.1 Loop-Based Implementation

In the loop-based implementation, gradient descent is performed by explicitly iterating over the dataset. The gradient computation involves manual summation across training examples, making the process computationally expensive.

4.2 Vectorized Implementation

The vectorized implementation uses NumPy matrix operations to compute predictions and gradients simultaneously for all data points. This removes explicit Python loops and leverages optimized linear algebra routines.

4.3 Training Procedure

Both models are trained using:

- Same initialization strategy for parameters
- Same learning rate
- Same number of iterations
- Identical data preprocessing pipeline

This ensures a fair comparison between both approaches.

5 Evaluation Metrics

The following evaluation metrics are used:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R-squared Score (R^2)
- Training Time (seconds)

6 Results

6.1 Cost Function Convergence

The convergence behavior of both implementations is shown below.

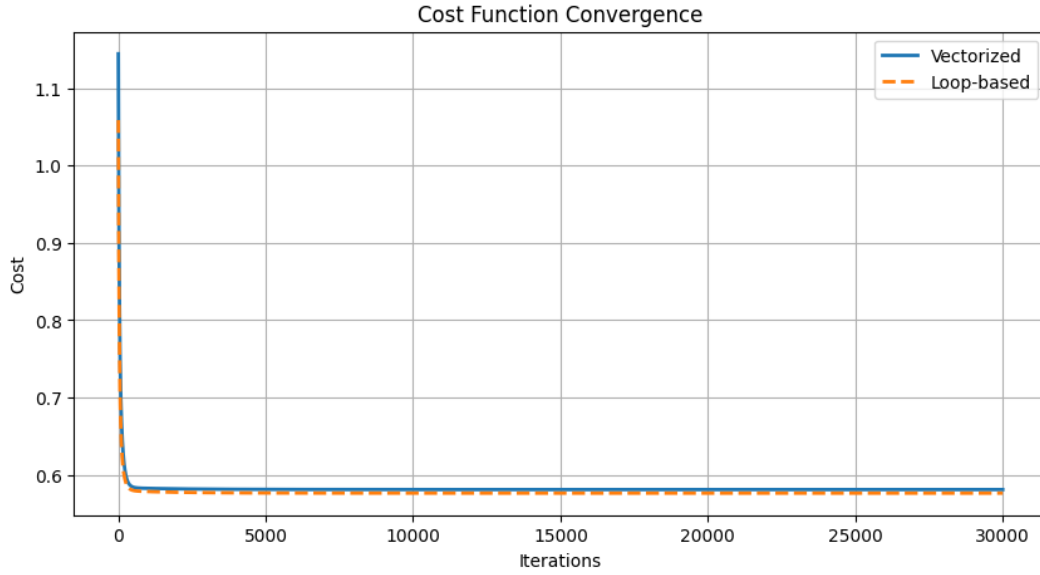


Figure 1: Cost function convergence over iterations for loop-based and vectorized implementations

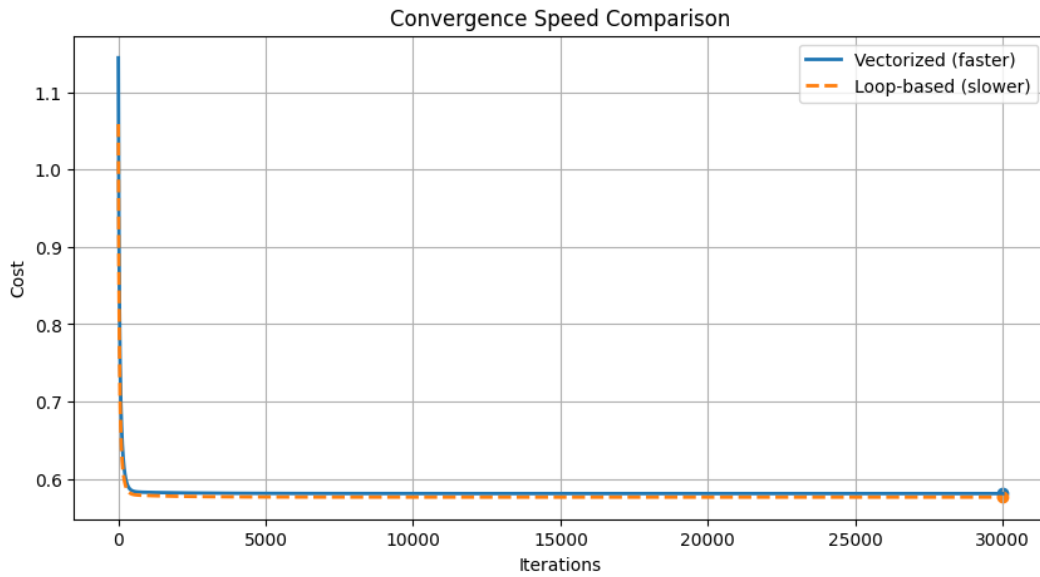


Figure 2: Comparison of convergence speed and final cost values between both implementations

6.2 Regression Performance Metrics

The predictive performance of both models is evaluated using MAE, RMSE, and R^2 score.

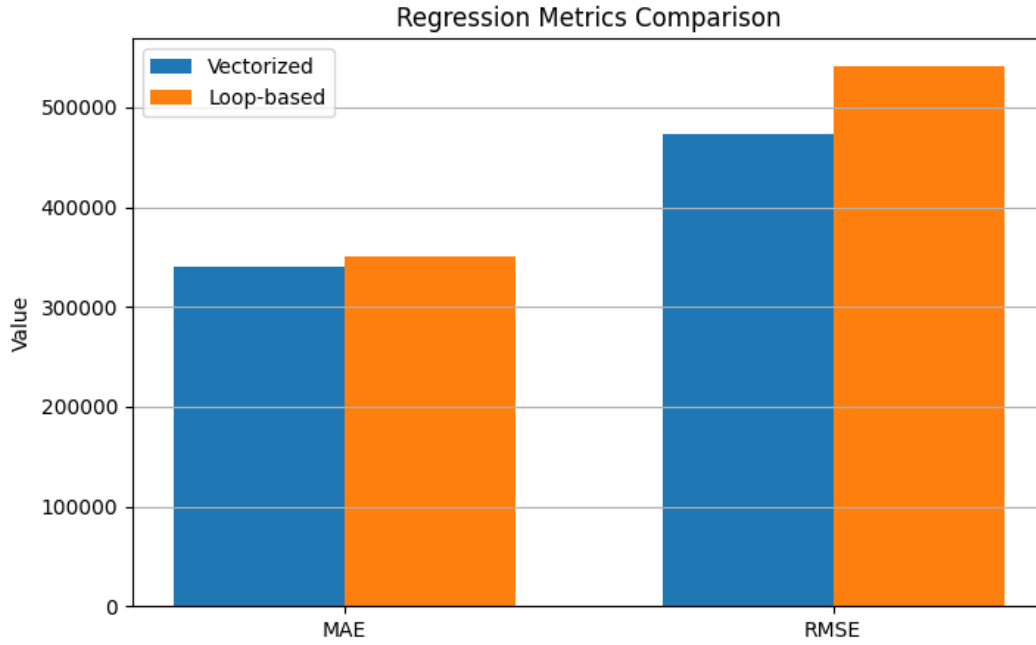


Figure 3: Regression performance comparison (MAE, RMSE, R^2)

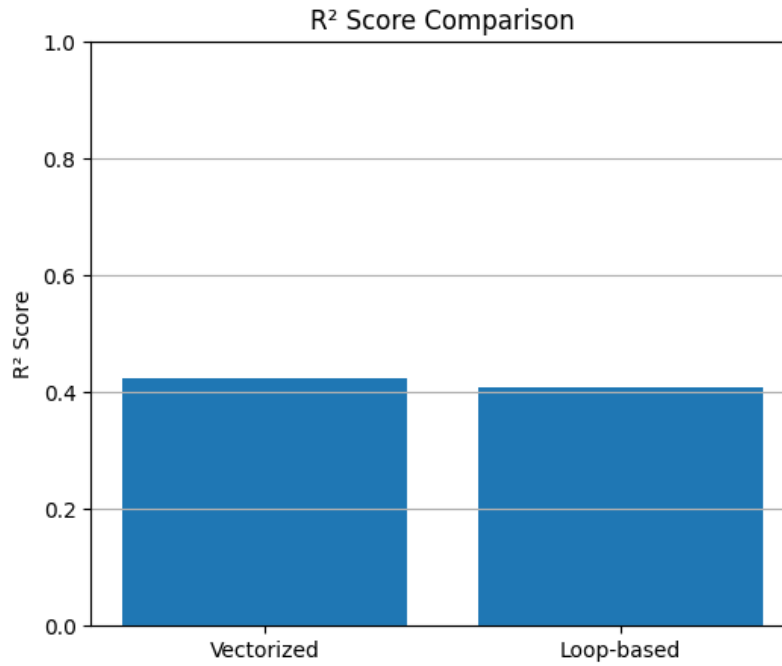


Figure 4: Regression performance comparison (MAE, RMSE, R^2)

7 Training Time Analysis

Training time is used as a measure of computational efficiency.

The vectorized implementation completes training significantly faster due to optimized matrix operations, while the loop-based implementation suffers from high computational overhead due to explicit iteration.

8 Discussion

The results clearly demonstrate that vectorization provides a substantial improvement in computational efficiency. The training time for the vectorized model is drastically lower compared to the loop-based implementation.

In terms of predictive performance, both models produce comparable results. However, the vectorized model slightly outperforms the loop-based approach in all evaluation metrics, indicating improved numerical efficiency and stability.

The convergence plots also show that both models follow a similar optimization path, but the vectorized version reaches the optimal region more efficiently.

9 Conclusion

This study highlights the importance of vectorization in machine learning implementations. While both loop-based and vectorized approaches converge to similar solutions, vectorization provides significant advantages in computational efficiency and slightly improved predictive performance.

Therefore, vectorized implementations are strongly recommended for practical machine learning applications, especially when dealing with large-scale datasets.